

STT 861: Theory of Probability & Statistics I

Lecture 11: Expected Values & Jensen's Inequality

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Casella & Berger, §2.2, pp. 55–58

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Motivation — What Is an Expected Value?

Intuition

The **expected value** of X is its *probability-weighted average* — the long-run average value if the experiment were repeated many times.

Fair die example. Roll a fair die: $X \in \{1, 2, 3, 4, 5, 6\}$, each with prob $1/6$.

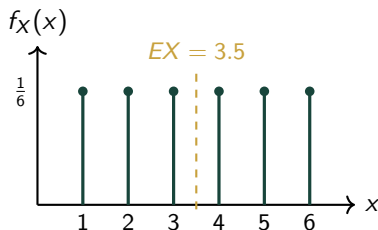
$$EX = \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = \frac{21}{6} = 3.5$$

You never actually roll a 3.5 — but over many rolls the average converges to 3.5.

Biased example. Suppose $P(X = 1) = 0.9$, $P(X = 10) = 0.1$.

$$EX = 0.9 \times 1 + 0.1 \times 10 = 0.9 + 1.0 = 1.9$$

The rare large value *pulls* the mean up.



Definition 2.2.1 — Expected Value / Mean

Definition 2.2.1 (CB)

The **expected value** (or **mean**) of a random variable $g(X)$, denoted $E g(X)$, is:

$$E g(X) = \begin{cases} \int_{-\infty}^{\infty} g(x) f_X(x) dx & \text{if } X \text{ is continuous} \\ \sum_{x \in \mathcal{X}} g(x) f_X(x) = \sum_{x \in \mathcal{X}} g(x) P(X = x) & \text{if } X \text{ is discrete} \end{cases}$$

provided the integral (or sum) exists. If $E|g(X)| = \infty$, we say $E g(X)$ **does not exist**.

Special cases

- $g(x) = x$: $EX =$ the **mean** of X , also written μ .
- $g(x) = x^2$: $EX^2 =$ the **second moment**.
- $g(x) = (x - \mu)^2$: $E(X - \mu)^2 =$ the **variance**.
- $g(x) = c$ (constant): $Ec = c$.

Two ways to compute $Eg(X)$

Method 1 (direct): use the definition with f_X .

$$Eg(X) = \int g(x) f_X(x) dx$$

Method 2 (transform): find f_Y where $Y = g(X)$, then

$$Eg(X) = EY = \int y f_Y(y) dy$$

Both give the same answer (Law of the Unconscious Statistician).

Example 2.2.2 — Exponential Mean

Example 2.2.2 (CB): Exponential(λ)

$$f_X(x) = \frac{1}{\lambda} e^{-x/\lambda}, \quad x > 0, \quad \lambda > 0.$$

$$EX = \int_0^{\infty} x \cdot \frac{1}{\lambda} e^{-x/\lambda} dx$$

Integrate by parts: $u = x$, $dv = \frac{1}{\lambda} e^{-x/\lambda} dx$:

$$\begin{aligned} EX &= \left[-x e^{-x/\lambda} \right]_0^{\infty} + \int_0^{\infty} e^{-x/\lambda} dx \\ &= 0 + \left[-\lambda e^{-x/\lambda} \right]_0^{\infty} = \lambda \end{aligned}$$

Interpretation

For an $\text{Exponential}(\lambda)$ random variable (e.g. waiting time), the mean waiting time is λ . The parameter λ serves double duty: it appears in both the pdf *and* equals the mean. (Some texts parameterise by $\beta = 1/\lambda$, giving $f_X(x) = \beta e^{-\beta x}$ with mean $1/\beta$.)

Example 2.2.3 — Binomial Mean

Example 2.2.3 (CB): Binomial(n, p)

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}, \quad x = 0, 1, \dots, n.$$

$$EX = \sum_{x=0}^n x \binom{n}{x} p^x (1 - p)^{n-x} = \sum_{x=1}^n x \binom{n}{x} p^x (1 - p)^{n-x}$$

Use the identity $x \binom{n}{x} = n \binom{n-1}{x-1}$:

$$\begin{aligned} EX &= \sum_{x=1}^n n \binom{n-1}{x-1} p^x (1 - p)^{n-x} \\ &= np \sum_{y=0}^{n-1} \binom{n-1}{y} p^y (1 - p)^{n-1-y} \quad (\text{set } y = x - 1) \\ &= np \cdot 1 = \mathbf{np} \quad (\text{The sum is the total of a Binomial}(n-1, p) \text{ pmf} = 1.) \end{aligned}$$

Example 2.2.4 — Cauchy Mean Does Not Exist

Example 2.2.4 (CB): Cauchy Distribution

$$f_X(x) = \frac{1}{\pi} \frac{1}{1+x^2}, \quad -\infty < x < \infty.$$

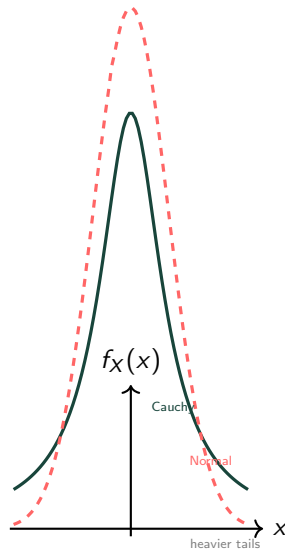
Does EX exist? Check $E|X|$:

$$E|X| = \int_{-\infty}^{\infty} |x| \cdot \frac{1}{\pi(1+x^2)} dx = \frac{2}{\pi} \int_0^{\infty} \frac{x}{1+x^2} dx = \frac{1}{\pi} \lim_{M \rightarrow \infty} \log(1+M^2) = \infty$$

Conclusion

$E|X| = \infty$, so EX **does not exist**.

The Cauchy distribution is the classic example of a distribution with *no mean*. Its heavy tails prevent the integral from converging.



Linearity: $E(aX + b) = aEX + b$

Linearity of Expectation (CB Eq. 2.2.1)

For any constants a and b :

$$E(aX + b) = aEX + b$$

Proof (continuous case):

$$\begin{aligned} E(aX + b) &= \int_{-\infty}^{\infty} (ax + b)f_X(x) dx \\ &= a \int x f_X(x) dx + b \int f_X(x) dx \\ &= aEX + b \cdot 1 = aEX + b. \quad \square \end{aligned}$$

Applications

(i) If $X \sim \text{Binomial}(n, p)$ with $EX = np$:

$$E(X - np) = np - np = 0$$

Centring at the mean gives zero expectation.

(ii) Temperature conversion: if $C \sim \text{some distribution}$ with $EC = 20^\circ$:

$$EF = E\left(\frac{9}{5}C + 32\right) = \frac{9}{5}(20) + 32 = 68^\circ$$

Theorem 2.2.5 — Properties of Expectation

Theorem 2.2.5 (CB)

Let X be a random variable and a, b, c constants. For any functions $g_1(x)$ and $g_2(x)$ whose expectations exist:

- Ⓐ $E(ag_1(X) + bg_2(X) + c) = aEg_1(X) + bEg_2(X) + c$ (**Linearity**)
- Ⓑ If $g_1(x) \geq 0$ for all x , then $Eg_1(X) \geq 0$ (**Nonnegativity**)
- Ⓒ If $g_1(x) \geq g_2(x)$ for all x , then $Eg_1(X) \geq Eg_2(X)$ (**Monotonicity**)
- Ⓓ If $a \leq g_1(x) \leq b$ for all x , then $a \leq Eg_1(X) \leq b$ (**Boundedness**)

Why these matter

- (a) allows us to compute $EX^2 = E[(X - \mu)^2] + (EX)^2$ and related identities without re-integrating.
- (b) and (c) give intuitive ordering: if X is always positive, so is its mean.
- (d) bounds the mean between the range of the function.

Example 2.2.6 — EX Minimises Mean Squared Error

Example 2.2.6 (CB): Minimising Distance

We want the constant b that best predicts X in the mean-squared sense. Minimise $E(X - b)^2$ over b .

Add and subtract $\mu = EX$:

$$\begin{aligned} E(X - b)^2 &= E[(X - \mu) + (\mu - b)]^2 \\ &= E(X - \mu)^2 + 2(\mu - b) \underbrace{E(X - \mu)}_{=0} + (\mu - b)^2 \\ &= E(X - \mu)^2 + (\mu - b)^2 \end{aligned}$$

The first term is fixed. The second term $(\mu - b)^2 \geq 0$ is minimised at $b = \mu$.

$$\min_b E(X - b)^2 = E(X - \mu)^2$$

Interpretation

EX is the **best constant predictor** of X under squared-error loss. This is the fundamental justification for using the mean as a measure of center.

Two Methods for $Eg(X)$

Two equivalent approaches

Method 1 (Direct): Use f_X directly:

$$Eg(X) = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

Method 2 (Transform): Find f_Y where $Y = g(X)$, then:

$$Eg(X) = EY = \int_{-\infty}^{\infty} y f_Y(y) dy$$

Both methods always give the same answer.

Example 2.2.7 (CB): Uniform-Exponential II

$X \sim \text{Uniform}(0, 1)$, $g(X) = -\log X$.

Method 1: $E(-\log X) = \int_0^1 (-\log x) \cdot 1 \, dx = [x - x \log x]_0^1 = 1$.

Method 2: From Ex 2.1.4, $Y = -\log X \sim \text{Exp}(1)$, so $EY = 1$ (by Ex 2.2.2 with $\lambda = 1$).

Both give $E(-\log X) = 1$. ✓

Additional Example A — Uniform Mean and EX^2

Example A: $X \sim \text{Uniform}(a, b)$

$$f_X(x) = \frac{1}{b-a}, \quad a < x < b.$$

Mean:

$$EX = \int_a^b x \cdot \frac{1}{b-a} dx = \frac{1}{b-a} \cdot \frac{b^2 - a^2}{2} = \frac{a+b}{2}$$

The mean is the *midpoint* of the interval — as expected by symmetry.

Second moment:

$$EX^2 = \int_a^b x^2 \cdot \frac{1}{b-a} dx = \frac{1}{b-a} \cdot \frac{b^3 - a^3}{3} = \frac{a^2 + ab + b^2}{3}$$

(Use $b^3 - a^3 = (b-a)(b^2 + ab + a^2)$.)

Additional Example B — Geometric Mean

Example B: $X \sim \text{Geometric}(p)$

$$P(X = x) = (1 - p)^{x-1}p, \quad x = 1, 2, 3, \dots$$

$$EX = \sum_{x=1}^{\infty} x(1 - p)^{x-1}p = p \sum_{x=1}^{\infty} x(1 - p)^{x-1}$$

Use the identity $\sum_{x=1}^{\infty} xr^{x-1} = \frac{1}{(1 - r)^2}$ for $|r| < 1$:

$$EX = p \cdot \frac{1}{(1 - (1 - p))^2} = p \cdot \frac{1}{p^2} = \frac{1}{p}$$

Interpretation

If each trial succeeds with probability p , on average you need $1/p$ trials to get the first success. E.g. fair coin ($p = 1/2$): expect 2 tosses to get the first head. Rare event ($p = 0.01$): expect 100 trials.

Additional Example C — Jensen's Inequality (Preview)

Jensen's Inequality (CB §2.2)

If g is a **convex** function ($g'' \geq 0$), then

$$Eg(X) \geq g(EX)$$

If g is **concave** ($g'' \leq 0$), the inequality reverses.

Applications

(a) $g(x) = x^2$ (convex): $EX^2 \geq (EX)^2$.

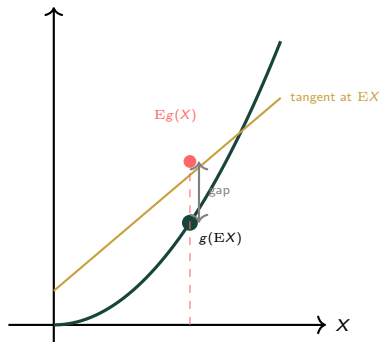
Rearranging: $\text{Var}(X) = EX^2 - (EX)^2 \geq 0$. ✓

(b) $g(x) = \log x$ (concave, $x > 0$): $E \log X \leq \log EX$.

(Arithmetic mean $\geq$ geometric mean.)

(c) $g(x) = e^x$ (convex): $Ee^X \geq e^{EX}$.

Used in moment generating functions.



Section 2.2 — Summary

Concept	Formula	Key result
Definition	$Eg(X) = \int g(x)f_X(x)dx$	Exists iff $E g(X) < \infty$
Linearity	$E(aX + b) = aEX + b$	Thm 2.2.5(a)
Exp(λ) mean	$EX = \lambda$	Ex 2.2.2
Binomial mean	$EX = np$	Ex 2.2.3
Cauchy mean	does not exist	Ex 2.2.4
Best predictor	$\min_b E(X - b)^2$ at $b = EX$	Ex 2.2.6
Jensen	convex g : $Eg(X) \geq g(EX)$	§2.2

Two methods for $Eg(X)$

Direct: $\int g(x)f_X(x) dx$ vs. **Transform:** find f_Y then $\int y f_Y(y) dy$.

Both give the same answer. Choose whichever integral is easier.

In-Class Activity — Section 2.2

Let X have pdf $f_X(x) = 3x^2$ on $[0, 1]$, zero elsewhere.

(a) Find EX . **(b)** Find EX^2 . **(c)** Find $E(3X - 1)$ using linearity (Thm 2.2.5a).

X has pmf $P(X = -1) = \frac{1}{4}$, $P(X = 0) = \frac{1}{2}$, $P(X = 2) = \frac{1}{4}$.

(a) Find EX . **(b)** Find EX^2 . **(c)** Find the constant b that minimises $E(X - b)^2$. Verify using Ex 2.2.6.

Let $X \sim \text{Uniform}(0, 1)$ and $g(X) = X^2$.

(a) Compute $Eg(X)$ directly using f_X .

(b) Find $f_Y(y)$ where $Y = X^2$ (from §2.1), then compute $EY = \int y f_Y(y) dy$.

(c) Confirm both methods agree.

Activity Solutions — Q1 & Q2

(a) $EX = \int_0^1 x \cdot 3x^2 dx = 3 \int_0^1 x^3 dx = 3 \cdot \frac{1}{4} = \frac{3}{4}$

(b) $EX^2 = \int_0^1 x^2 \cdot 3x^2 dx = 3 \int_0^1 x^4 dx = 3 \cdot \frac{1}{5} = \frac{3}{5}$

(c) By linearity: $E(3X - 1) = 3EX - 1 = 3 \cdot \frac{3}{4} - 1 = \frac{9}{4} - 1 = \frac{5}{4}$

(a) $EX = (-1) \cdot \frac{1}{4} + 0 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} = -\frac{1}{4} + 0 + \frac{2}{4} = \frac{1}{4}$

(b) $EX^2 = (-1)^2 \cdot \frac{1}{4} + 0^2 \cdot \frac{1}{2} + 2^2 \cdot \frac{1}{4} = \frac{1}{4} + 0 + \frac{4}{4} = \frac{5}{4}$

(c) By Example 2.2.6, $E(X - b)^2$ is minimised at $b = EX = \frac{1}{4}$. Minimum value = $E(X - \frac{1}{4})^2 = EX^2 - (EX)^2 = \frac{5}{4} - \frac{1}{16} = \frac{19}{16}$.

Activity Solutions — Q3 & Q4

(a) Direct: $EX^2 = \int_0^1 x^2 \cdot 1 \, dx = \frac{1}{3}.$

(b) From §2.1 Ex 2.1.7: $f_Y(y) = \frac{1}{2\sqrt{y}}, 0 < y < 1.$

$$EY = \int_0^1 y \cdot \frac{1}{2\sqrt{y}} \, dy = \int_0^1 \frac{\sqrt{y}}{2} \, dy = \frac{1}{2} \cdot \frac{2}{3} y^{3/2} \Big|_0^1 = \frac{1}{3}$$

(c) Both methods give $EX^2 = \frac{1}{3}.$ ✓ This confirms the **Law of the Unconscious Statistician**.

(a) $EX = \frac{1}{2}.$ $Ee^X = \int_0^1 e^x \, dx = [e^x]_0^1 = e - 1 \approx 1.718.$

(b) $e^{EX} = e^{1/2} = \sqrt{e} \approx 1.649.$

(c) $Ee^X = e - 1 \approx 1.718 > \sqrt{e} \approx 1.649 = e^{EX}.$ ✓

$g(x) = e^x$ is convex since $g''(x) = e^x > 0$, so Jensen gives $Ee^X \geq e^{EX}.$